Von Karman plates, existence results and numerical tests

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Abstract

We study thin plates where out of plane deformations are activated by wedge disclinations which are configurations observed in metal plates and graphene. ...We are interested in the situation where out-of-plane deflections are activated by membrane kinematic incompatibility, noticeably, Volterra's disclinations. We show existence of equilibrium solutions in the presence of a finite system of disclinations and a dead load for the deflection under various boundary conditions. We set up a numerical code in Fenics environment for the resolution of the equilibrium equations which we study in a set of parametric test studies. Our implementation of the Finite Element method follows the Discontinuous Galerkin Method with C^0 elements.

Table 1: Material and geometric parameters.

Symbol	**Description**	**Units**
$s \in \mathbb{R}$	Frank angle	rad
$\mathsf{E} > 0$	Young modulus	Pa
$\nu \in [0, \frac{1}{2}]$	Poisson ratio	non-dimensional
$\mathsf{D} := \frac{\mathsf{E}}{1-\nu^2}\frac{h^3}{12}$	Plate's bending stiffness	N m
$h > 0$	Plate thickness	m
$\Omega \subset \mathbb{R}^2$	Plate's elastic surface in its reference configuration	N.A.
$p : \Omega \to \mathbb{R}$	Transverse load per unit area	nondim
$p_0 \in \mathbb{R}$	p's scale	Pa
$R > 0$	Plate's characteristic length	m
$x = (x_1, x_2) \in \Omega$	Coordinate system	m
$w : \Omega \to \mathbb{R}$	Transverse displacement	nondim
$w_0 > 0$	w scale	m
$v : \Omega \to \mathbb{R}$	Airy's stress function	nondim
$v_0 > 0$	v scale	N m
$\mathcal{E}_{\rm bd}$	Boundary edges	N.A.
$\mathcal{E}_{\rm in}$	Inner edges	N.A.
$[u]$	Average of u over ... (TOFIX$^{\rm ALB}$)	nondim
v	Discontinuous Galerkin approximation of v	nondim
w	Discontinuous Galerkin approximation of w	nondim
φ	Test function	nondim

0.1 Discrete Numerical Formulation

- notation: $\mathcal{F}$ macro
- parentheses for arguments from square to round
- tildas on domain
- notation: check the table

We adopt C^0-elements with an interior penalty approach to weakly enforce gradient continuity across interior mesh interfaces (see [?]). This results into a non-conforming Discontinuous Galerkin (DG) method [?, ?] for our fourth-order problem formulated in H^2. A comprehensive comparison of DG methods for fourth-order problems can be found in [?], while standard implementations for biharmonic problems are discussed in [?, ?]. For the specific case of von Kármán equations, C^0-DG methods and corresponding error estimates are provided in [?]. Our implementation builds upon the formulation in [?], but instead of embedding the penalty terms into the discrete weak form, we generalise the approach by incorporating them directly into the energy functional. ALB

[ALB 1] conforming?

Discontinuous Galerkin methods are non-conforming finite element methods, which rules out restricting the energy functional $\widetilde{\mathcal{F}}$ to a finite-dimensional subspace of $H_0^2(\Omega) \times H_0^2(\Omega)$ to derive the discrete weak formulation of problem (18). Following [?], a proper DG formulation requires adding additional terms to the energy functional. These terms account for discontinuities in the gradient across mesh interfaces and ensure the coercivity of the problem. Let $\mathcal{T}$ denote a regular triangulation of $\Omega \subset \mathbb{R}^2$. Define $\mathcal{E}_{\text{bd}}$ as the set of boundary edges, $\mathcal{E}_{\text{in}}$ as the set of internal edges, so that $\mathcal{E} := \mathcal{E}_{\text{in}} \cup \mathcal{E}_{\text{bd}}$ is the collection of all edges. For $T \in \mathcal{T}$, let $\mathrm{P}_3(T)$ be the space of polynomials of degree at most three defined on T. The Lagrange finite element space $\mathcal{V}(\mathcal{T})$ consists of globally continuous, piecewise cubic polynomials that vanish on the boundary edges $\mathcal{E}_{\text{bd}}$, namely ALB

$$\mathcal{V}(\mathcal{T}) := \left\{ u \in C^0(\Omega) : u|_T \in \mathrm{P}_3(T) \quad \forall T \in \mathcal{T} \right\} \cap H_0^1(\Omega) \tag{1}$$

[ALB 2] unclear def. $\pm$ if T on the boundary

For any two triangles T^+ and T^- such that $T^+ \cap T^- \in \mathcal{E}_{\text{in}}$ and for any triangle $T^\pm \in \mathcal{E}_{\text{bd}}$, the jump and average operators are defined as follows

$$[\![u]\!]_e := \begin{cases} u|_{T^+} - u|_{T^-} & \text{if } e \in \mathcal{E}_{\text{in}} \\ u|_{T^\pm} & \text{if } e \in \mathcal{E}_{\text{bd}} \end{cases} \qquad [u]_e := \begin{cases} \frac{u|_{T^+} + u|_{T^-}}{2} & \text{if } e \in \mathcal{E}_{\text{in}} \\ u|_{T^\pm} & \text{if } e \in \mathcal{E}_{\text{bd}} \end{cases} \tag{2}$$

For any $e \in \mathcal{E}$, denote by $t_e := \operatorname{diam}(e)$ the diameter of the element. We conventionally define n_e, the outward unit normal on $e \in \mathcal{E}_{\text{in}}$ pointing from T^- to T^+. Note that $[\![\partial_{n_e} u]\!]$ and $\{\partial_{n_e n_e} u\}$ are independent on the particular labelling of two adjacent triangles, and definition (2) is applied component-wise when u is a vector-valued or a matrix-valued function. Based on the definition (??) and on Remark ?? we define

$$\mathcal{F} : \mathcal{V}(\mathcal{T}) \times \mathcal{V}(\mathcal{T}) \mapsto \mathbb{R} \quad \text{such that} \quad (v, w) \mapsto \mathcal{F}(v, w) \tag{3}$$

and

$$L_j : \mathcal{V}(\mathcal{T}) \mapsto \mathbb{R} \quad \text{such that} \quad u \mapsto L_j(u) \quad \text{for } j = \{1, 2\}, \tag{4}$$

where

[ALB 3] is K the triangle

$$\mathcal{F}(v,w) :=$$
$$\sum_{T\in\mathcal{T}}\left(-\frac{1}{2}\|\Delta v\|^2_{L^2(T\underline{K}^{\mathrm{ALB}})}+\frac{1}{2}\int_{T\underline{K}^{\mathrm{ALB}}}\mathrm{cof}(\nabla^2 v):(\nabla w\otimes\nabla w)d\xi+\frac{c_\nu}{2}\|\Delta w\|^2_{L^2(T\underline{K}^{\mathrm{ALB}})}\right)$$
$$-\sum_{T\in\mathcal{T}}\gamma\beta^2\int_{T\underline{K}^{\mathrm{ALB}}}wpd\xi+\beta^2\langle v,\theta\rangle, \quad (5)$$

$$L_1(u) := -\frac{1}{2}\sum_{e\in\mathcal{E}}\int_e[\![\nabla u\cdot n_e]\!]_e[\nabla^2 u n_e\cdot n_e]_e d\mathcal{H}^1, \quad L_2(u) := \frac{1}{2}\sum_{e\in\mathcal{E}}\frac{\alpha}{[t_e]}\int_e[\![\nabla u\cdot n_e]\!]^2_e d\mathcal{H}^1. \quad (6)$$

When differentiated, L_1 gives rise to two terms: one naturally emerges when the energy functional $\widetilde{\mathcal{F}}$ is embedded in ~~larger~~^{ALB} a space where the continuity of the gradient is not guaranteed, while the second ensures symmetry~~added to make the IPDG formulation symmetric~~^{ALB} (see [?] for details). On the other hand, L_2 introduces a stabilization term which ensures coercivity, provided the interior penalty parameter α is positive and sufficiently large^{ALB} (see [?]).

[ALB 4] in which natural sense?

We are now in a position to establish a symmetric variational C^0-DG formulation of problem (18) (with interior penalty). Define

$$\mathcal{I}:\mathcal{V}(\mathcal{T})\times\mathcal{V}(\mathcal{T})\mapsto\mathbb{R} \quad \text{such that} \quad (v,w)\mapsto\mathcal{I}(v,w) \quad (7)$$

[ALB 5] unclear

where

$$\mathcal{I}(v,w) := \mathcal{F}(v,w)+\frac{c_\nu}{2}L_1(w)+\frac{c_\nu}{2}L_2(w)-\frac{1}{2}L_1(v)-\frac{1}{2}L_2(v)$$
$$-\frac{1}{2}\sum_{e\in\mathcal{E}_{\mathrm{in}}}\int_e[\mathrm{cof}(\nabla w\otimes\nabla w):n_e\otimes n_e]_e[\![\nabla v\cdot n_e]\!]_e d\mathcal{H}^1 \quad (8)$$

where the last term is taken from the formulation developed in [?]. ~~Now, we may formulate the Symmetric IPC^{0}DG variational formulation as.~~
^{ALB} We seek $(v,w)\in\mathcal{V}(\mathcal{T})\times\mathcal{V}(\mathcal{T})$ such that^{ALB}

$$\lim_{\epsilon\to0}\frac{\mathcal{I}(v+\epsilon\varphi,w)}{d\epsilon}\Big|_{\epsilon=0}=0 \quad \forall\varphi\in\mathcal{V}(\mathcal{T}) \quad (9)$$
$$\lim_{\epsilon\to0}\frac{\mathcal{I}(v,w+\epsilon\varphi)}{d\epsilon}\Big|_{\epsilon=0}=0 \quad \forall\varphi\in\mathcal{V}(\mathcal{T}) \quad (10)$$

To assess the performance of our approach we also present two alternative formulations available in the literature.~~For the sake of completeness, we also report two other IPC^{0}DG formulation available in the literature due to, respectively, [?] and [?]. We will use them in the next section to asses the performance of our formulation. The IPC^{0}DG Finite Element formulation due to [?] reads:~~^{ALB} The first, proposed by [?]: find $(v,w)\in\mathcal{V}(\mathcal{T})\times\mathcal{V}(\mathcal{T})$ such that^{ALB}

$$-\sum_{T\in\mathcal{T}}\Big(\int_T\nabla^2 v:\nabla^2\varphi dx-\frac{1}{2}\int_T[w,w]\varphi dx\Big)+\sum_{e\in\mathcal{E}}\Big(+\int_e[\![\nabla v\cdot n_e]\!]_e[\nabla^2\phi n_e\cdot n_e]_e d\mathcal{H}^1+$$
$$+\int_e[\![\nabla\phi\cdot n_e]\!]_e[\nabla^2 v n_e\cdot n_e]_e d\mathcal{H}^1-\frac{\alpha}{[t_e]}\int_e[\![\nabla v\cdot n_e]\!]_e[\![\nabla\phi\cdot n_e]\!]_e d\mathcal{H}^1+$$
$$\underbrace{\frac{1}{2}\int_e[\![[\mathrm{cof}(\nabla^2 w)]_e\nabla w\cdot n_e]\!]_e\varphi d\mathcal{H}^1}_{\bigstar}\Big)=-\beta^2\langle\varphi,\tilde{\theta}\rangle \quad \forall\varphi\in\mathcal{V}(\mathcal{T}) \quad (11)$$

and

$$\sum_{T\in\mathcal{T}}\Big(\int_T \nabla^2 w : \nabla^2\varphi dx - \int_T [v,w]\varphi dx + \sum_{e\in\mathcal{E}}\Big(+ \int_e [\![\nabla w\cdot n_e]\!]_e [\nabla^2\varphi n_e\cdot n_e]_e d\mathcal{H}^1 + \\ + \int_e [\![\nabla\varphi\cdot n_e]\!]_e [\nabla^2 w n_e\cdot n_e]_e d\mathcal{H}^1 + + \frac{\alpha}{[t_e]}\int_e [\![\nabla w\cdot n_e]\!]_e [\![\nabla\varphi\cdot n_e]\!]_e d\mathcal{H}^1 + \\ \underbrace{+\frac{1}{2}\int_e \left([\![[\mathrm{cof}(\nabla^2 v)]_e\nabla w\cdot n_e]\!]_e + [\![[\mathrm{cof}(\nabla^2 w)]_e\nabla v\cdot n_e]\!]_e\right)\psi d\mathcal{H}^1}_{\bigstar} = \\ = \gamma\beta^2\sum_{T\in\mathcal{T}}\int_T \varphi p dx \quad \forall\varphi\in\mathcal{V}(\mathcal{T}). \quad (12)$$

And the second, proposed by [?]: find $(v,w)\in\mathcal{V}(\mathcal{T})\times\mathcal{V}(\mathcal{T})$ such that (v,w) satisfy (11) and (12) where the terms marked with ★ are omitted. ~~whereas the IPC^{0}DG Finite Element formulation due to [?] can be obtained omitting the term $\frac{1}{2}\int_e [\![[\mathrm{cof}(\nabla^2 w)]_e\nabla w\cdot n_e]\!]_e\varphi d\mathcal{H}^1$ in the Eqn. (11) and the terms $\int_e [\![[\mathrm{cof}(\nabla^2 v)]_e\nabla w\cdot n_e]\!]_e\varphi d\mathcal{H}^1$ and $\int_e [\![[\mathrm{cof}(\nabla^2 w)]_e\nabla v\cdot n_e]\!]_e\varphi d\mathcal{H}^1$ in the Eqn. (12).~~ ALB

0.2 Non-dimensional formulation

To highlight the role of key parameters, we introduce a non-dimensional formulation by rescaling dependent and independent variables as follows:

$$\xi := x/R, \qquad \tilde{p}(\xi) := p(x)/p_0, \qquad \tilde{w}(\xi) := w(x)/w_0, \qquad \tilde{v}(\xi) := w(x)/v_0, \qquad (13)$$

where p_0, w_0, and v_0 are arbitrary scaling factors to fix. As a consequence of the rescaling of spatial coordinates, we denote by $\widetilde{\Omega}$ the domain Ω rescaled by its characteristic length R, and by $[\cdot,\cdot]_\xi := R^2[\cdot,\cdot]$ and $\Delta_\xi\cdot := R^2\Delta\cdot$, the scaled Monge-Ampère bracket and Laplacian operator, respectively. Plugging in (**??**) we readily obtain

$$\mathcal{F}(v,w) = -\frac{v_0^2}{EhR^2}\frac{1}{2}\int_{\widetilde{\Omega}}(\Delta_\xi\tilde{v})^2\,d\xi + \frac{Eh^3w_0^2}{R^2}\frac{c_\nu}{2}\int_{\widetilde{\Omega}}(\Delta_\xi\tilde{w})^2\,d\xi - \frac{v_0w_0^2}{2R^2}\int_{\widetilde{\Omega}}[\tilde{w},\tilde{w}]_\xi\,\tilde{v}\,d\xi + \\ - R^2p_0w_0\int_\Omega \tilde{p}\,\tilde{w}\,d\xi + v_0\langle\tilde{\theta},\tilde{v}\rangle =: F(\tilde{v},\tilde{w}) \quad (14)$$

where $\tilde{\theta} := \sum_{i=1}^N s_i\delta(\tau_i-\xi)$, $\tau_i := t_i/R$ for every $i\in\{1,\dots,N\}$, and $c_\nu := 1/(12(1-\nu^2))$. Rescaling $\mathcal{F}$ by the positive quantity $\frac{v_0^2}{EhR^2}$, we associate to pairs $(\tilde{v},\tilde{w})\in H_0^2(\widetilde{\Omega})\times H_0^2(\widetilde{\Omega})$ the following energy functional

$$\widetilde{\mathcal{F}}(\tilde{v},\tilde{w}) := -\frac{1}{2}\int_{\widetilde{\Omega}}(\Delta_\xi\tilde{v})^2\,d\xi + \frac{E^2h^4w_o^2}{v_o^2}\frac{c_\nu}{2}\int_{\widetilde{\Omega}}(\Delta_\xi w)^2\,d\xi - \frac{Ehw_o^2}{v_o}\frac{1}{2}\int_{\widetilde{\Omega}}[\tilde{w},\tilde{w}]_\xi\,\tilde{v}\,d\xi + \\ - \frac{p_ow_oER^4h}{v_o^2}\int_{\widetilde{\Omega}}\tilde{p}\,\tilde{w}\,d\xi + \frac{ER^2h}{v_o}\langle\tilde{\theta},\tilde{v}\rangle \quad (15)$$

Since the scales are arbitrary, we choose w_0 and v_0 such that the coefficients of the bending and coupling energy terms in (15) are equal to 1. Solving the system

$$E^2h^4w_0^2 = v_0^2, \qquad Ehw_0^2 = v_0 \qquad (16)$$

ALB we identify $w_0 = h$ and $v_0 = Eh^3$, which in turn implies that $\mathcal{F}$ globally scales as Eh^5/R^2. Dropping the tildas for ease of notation, the non-dimensional energy finally reads

$$\mathcal{F}(v,w) := -\frac{1}{2}\int_\Omega (\Delta_\xi v)^2\, d\xi + \frac{c_\nu}{2}\int_\Omega (\Delta_\xi w)^2\, d\xi - \frac{1}{2}\int_\Omega [w,w]_\xi\, v\, d\xi - \gamma\beta^4 \int_\Omega p\, w d\xi + \beta^2\langle\theta, v\rangle \tag{17}$$

which features two non-dimensional parameters, namely the plate's geometric aspect ratio $\beta := R/h$ and the scaled pressure $\gamma := p_0/E$, measuring the transverse load relative to the (three-dimensional) material stiffness. Similarly, the non-dimensional form of Eqn (**??**) reads

$$\begin{cases} c_\nu \Delta_\xi^2 w = [v,w]_\xi + \gamma p & \text{in } \Omega, \\ \Delta_\xi^2 v = -\frac{1}{2}[w,w]_\xi + \beta^2\theta & \text{in } \Omega. \end{cases} \tag{18}$$

In the non-dimensional formulation, β^2 acts as a geometric gain factor for the distribution of disclinations, while γ scales the transverse external load.

To gauge physically relevant values for the non-dimensional parameters, consider a metallic thin plate of thickness $h = 0.01$m and radius $R = 1$m, with a Young's modulus $E = 10$GPa, and a Poisson ratio $\nu = 0.15$, subject to a transverse pressure due to a distributed load comparable to its own weight. Assuming a density $\rho \sim 10^3 kg/m^3$ (corresponding to a reference density for steel) and denoting by g the gravitational acceleration, estimating the characteristic pressure as $p \sim \rho g h \sim 10^2$Pa, we obtain the reference values

$$\overline{\beta} = 10^2, \qquad \overline{\gamma} = 10^{-8}. \tag{19}$$

The range of values for the aspect ratio β and the relative transverse pressure γ that we will consider in the following numerical experiments span the physical range of moderately thick ($\beta \sim 10$) to very thin ($\beta \sim X00$) plates, and transverse loads from negligible (1%) to 10-fold the plate's own weight.

1 Numerical experiments

Implementation We address the numerical solution of the variational problem (**??**) leveraging dolfinx/FEniCSx [**?**] as finite element framework and PETSc [**?**, **?**] for efficient and scalable linear algebra. Our implementation `footnote: https://github.com/kumiori/disclin` implements polynomial (Lagrange) finite elements whose order can be fixed at runtime, for flexibility in the discretisation of the variational principle. More specifically, we use SNES, the Newton nonlinear solver provided by PETSc. The iterative subproblems are solved by a semi-direct Krylov subspace method with LU factorization as the preconditioning step (ksp...: preonly, pc type: lu).ALB

Experiments are performed on a unit disc $\Omega := B_1(0)$ centered in the origin, meshed with triangular (Delaunay) elements of typical size $\eta = 0.05$, corresponding to 1,595 nodes and xxxx elements. We employ third-order Lagrange elements for both the transverse displacement and the Airy stress potential, providing a piecewise linear approximation of the Gaussian curvature. This choice of discretisation yields matrix sizes of approx. ...K nonzero entries. The non-dimensional formulation involves two tunable parameters: the aspect ratio ($\beta = R/h \gg 1$) measuring the plate's slenderness, and the relative transverse pressure (p_0/E) with respect to the Young modulus, measuring the membrane stiffness3d

Young[ALB]. Finally, the disclination strength given by the Frank angle, is fixed to ± 1, to qualitatively distinguish positive from negative induced membrane stresses, respectively corresponding to membrane tension and compression. [ALB] All computations throughout this section use the non-dimensional formulation, likewise the plots representing the energy contributions and the mechanical fields. [ALB]

[ALB 6] CHECK

1.1 Solver verification

Two verification experiments, functional in establishing the reliability of the numerical framework with respect to closed form solutions, substantiate that the computational implementation accurately represents the mathematical model within specified accuracy. We focus on two configurations: kinematically compatible loads by transverse surface pressure and kinematically incompatible loads generated by a disclination dipole, symmetric with respect to the centre of the plate. [ALB]

1.1.1 Test 1: Kinematically compatible plate under transverse pressure

Consider Eqn. (**??**) and assume $\theta = 0$ and $p(r) := \sqrt{\frac{2\mathsf{D}^3}{\mathsf{E}h}}\left(\frac{40}{3}(1-r^2)^4 + \frac{16}{3}(11+r^2)\right)$ where $r = |x|$ is the radial coordinate of a polar reference system. The pair (w^*, v^*) where

$$w^*(r) := \sqrt{\frac{2\mathsf{D}}{\mathsf{E}h}}(1-r^2)^2 \qquad v^*(r) := \frac{\mathsf{D}}{h}\left(-\frac{1}{12}(1-r^2)^2 - \frac{1}{18}(1-r^2)^3 - \frac{1}{24}(1-r^2)^4\right) \tag{20}$$

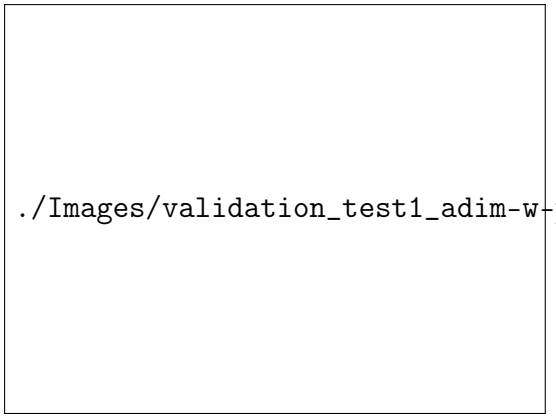
$$w^*(r) := w_0(1-r^2)^2 \qquad v^*(r) := v_0\left(-\frac{1}{12}(1-r^2)^2 - \frac{1}{18}(1-r^2)^3 - \frac{1}{24}(1-r^2)^4\right) \tag{21}$$

is a solution to Eqn. (**??**). In Figures 1-(a) and 1-(b) we show the numerical solutions obtained from the three discrete formulations (see section 0.1) and the pair (w^*, v^*), along the segment $x_2 = 0$. Here, the non-dimensional parameters read $p_0 = \sqrt{\frac{2D^3}{Eh}}$, $w_0 = \sqrt{\frac{2D}{Eh}}$, and $v_0 = \frac{D}{h}$. Note that the nondimensional formulation involving transverse loading only is independent of the aspect ratio. Remark that, because v plays the role of the membrane stress potential (?), its sign locally informs of the positive vs. negative character of membrane strains. Transverse pressure induces bending which in turn, induces tensile (positive) membrane strains. ... inversion of curvature due to the clamping boundary condition. In the image, the red locus corresponds to $x : \kappa(x) = 0$ [ALB] There are no significant deviations among the three finite element formulations. We show in table 2 the relative energy errors with respect to the energy of the exact solution.

1.1.2 Test 2: Kinematically Incompatible plate, disclination dipole

A circular plate loaded with a dipole of disclinations symmetrically arranged around the centre of the plate constitutes our second verification test.[ALB] Consider Eqn. (**??**) and assume $\theta = \delta(x - 0.2e_1)\ -\delta(x + 0.2e_1)$, $p(x) = 0$ where e_1 is the unit vector oriented according to the x_1-axis. The pair (w^*, v^*), given by

$$w^*(x) = 0 \qquad v^*(x) = \frac{Eh}{16\pi}\big(\mathcal{G}(x, -0.2) - \mathcal{G}(x, 0.2)\big) \tag{22}$$



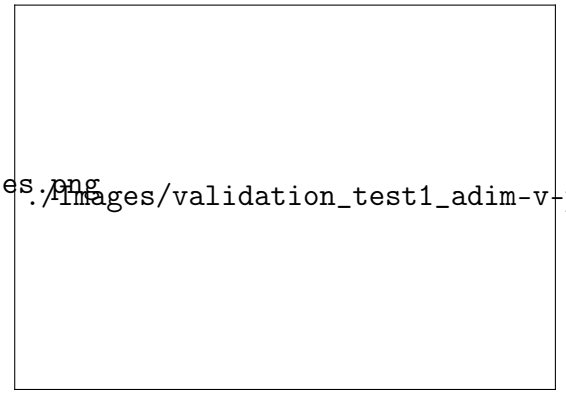


Figure 1: LEFT: $w(x_1, 0)$. RIGHT: $v(x_1, 0)$. $x_1 \in [-1, 1]$. Blue line: variational FE formulation. Dotted red line: [**?**] FE formulation. Green dashed line: [**?**] FE formulation. Dashed black line: analytical solution.

Energy [J]	**Exact Value**	e [VAR]	e [[**?**]]	e [[**?**]]
Bending	0.486	-0.13%	-0.07%	-0.06%
Membrane	0.006	-0.23%	-0.66%	-0.69%
Coupling	0.013	0.83%	-0.43%	-0.43%

Table 2: Comparison of exact energy values with relative errors across different methods.

where

$$\mathcal{G}(x,t) := \begin{cases} (1-t^2)(1-x_1^2-x_2^2) + \left((t-x_1)^2+x_2^2\right) \ln \frac{(t-x_1)^2+x_2^2}{t^2x_2^2+(tx_1-1)^2} & \text{for } (x_1,x_2) \neq (t,0) \\ 0 & \text{otherwise} \end{cases} \tag{23}$$

is a solution .... We solve [...] with $\gamma = 0$, for varying aspect ratio β between 10 and $X00$.[ALB] Figure 2 shows the profile of the dimensional Airy's stress function, comparing the three different finite element approximations to the exact solution. As expected, the transverse displacement is identically zero for all three discrete formulations. Figure, ... shows the energy dependence to the fourth power of the aspect ratio, β. This behaviour corresponds to the expected scaling, remarking that the multiplicative factor in the membrane equations - linear when $w = 0$ - is β^2. On the other hand, the bending energy and the coupling energy are zero. We compare in Figure the profiles of approximate finite element solutions with respect to the closed form analytic solutions, along the segment $x_2 = 0$. Figure ... shows the fields w, v on the two-dimensional computational domain, the dots mark the locations of the disclinations (colour coding), Figure shows the scalar stress potential associated to the positive (colour) and negative (colour) disclination, respectively inducing a locally tensile and compressive state. [ALB]

[ALB 7] dimensional?

The relative energy error for the membrane component is: -0.301% for the variational formulation, and -0.518% for both the the formulations described in [**?**] and [**?**]. The bending and coupling energies are zero in all the three FE formulations.

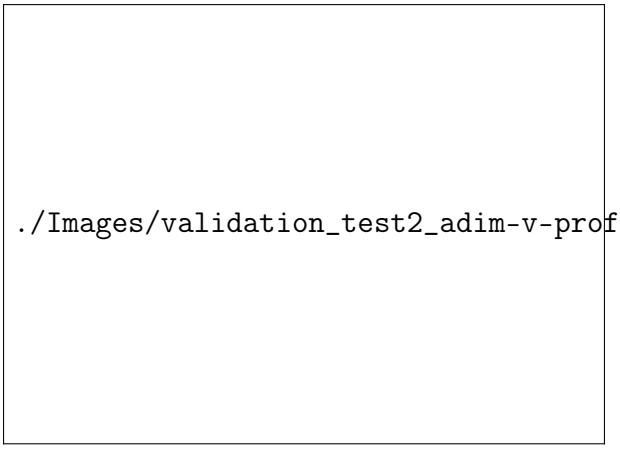


Figure 2: $v(x_1, 0)$ for $x_1 \in [-1, 1]$. Blue line: variational FE formulation. Dotted red line: [**?**] FE formulation. Green dashed line: [**?**] FE formulation. Dashed black line: analytical solution.

1.2 Parametric experiments

Mixed Loading: Disclination monopole and surface pressure, Varying Thickness We analyse in this section the mechanics of the nonlinear system by varying first β in some appropriate while keeping γ constant; and then varying γ with constant β. In order to do this, we provide an estimate for a realistic range of values for both the dimensionless parameters. The first series of parametric experiments investigating mixed-mode loading combine surface pressure and a disclination monopole, as the plate's aspect ratio is varied between 10 and $X00$ keeping constant the prefactor of the transverse external load $\gamma\beta^4 = 1$. As β varies, the physical dimensional surface pressure decreases with the inverse fourth power of the dimensional thickness. Nondimensional energy plots show the energy scaling of the different components of the energy. Similarly to the dipole setting, the membrane energy scales with the fourth power of the aspect ratio, whereas the nondimensional bending energy decreases with a exponent of -1.65, numerically computed by linear regression. For large values of the aspect ratio, small numerical oscillations appear on the bending term, whose order of magnitude several orders smaller than the dominant membrane energy term. Non-dimensional transverse displacement goes to zero.
ALB In this section we consider $\theta := -\delta(x)$.

[ALB 8] dimensional

The effect of the aspect ratio β We study now how different values of β affects the plate's equilibrium configuration when γ is equal to its reference value $\overline{\gamma}$. We consider values of β in the range $[10, 100]$, showing in Figure 3 how the dimensionless membrane, bending and coupling energies depend upon β. The membrane energy matches the (membrane) energy predicted by the linear Kirchhoff-Love plate model [**?**], namely $E_{\mathrm{KL}} = \frac{\beta^4}{32\pi}$, suggesting a partial decoupling of the governing equations. Indeed, Eqn. (18) can be written as two weakly coupled equations, where the membrane stress can is autonomous and enters as a datum in for the transverse equation, namely

[ALB 9] unclear

$$\Delta_\xi^2 \widetilde{v}(\xi) \approx -\beta^2 \delta(\xi) \quad \text{and} \quad c_\nu \Delta_\xi^2 \tilde{w} = [\tilde{v}, \tilde{w}]_\xi + f, \text{ in } \widetilde{\Omega} \tag{24}$$

Eqn. (24)-(1) can now be solved separately and its solution plugged into (24)-(2). Due to the XXXX sign of the disclination, $\tilde{v}$ induces a tensile stress field on the plate, as it is shown in Figure 4 where the radial stress $\sigma_{rr} = \sigma e_r \cdot e_r$ is everywhere non-negative. This

[ALB 10] drop the tildas

[ALB 11] rel. sigma-

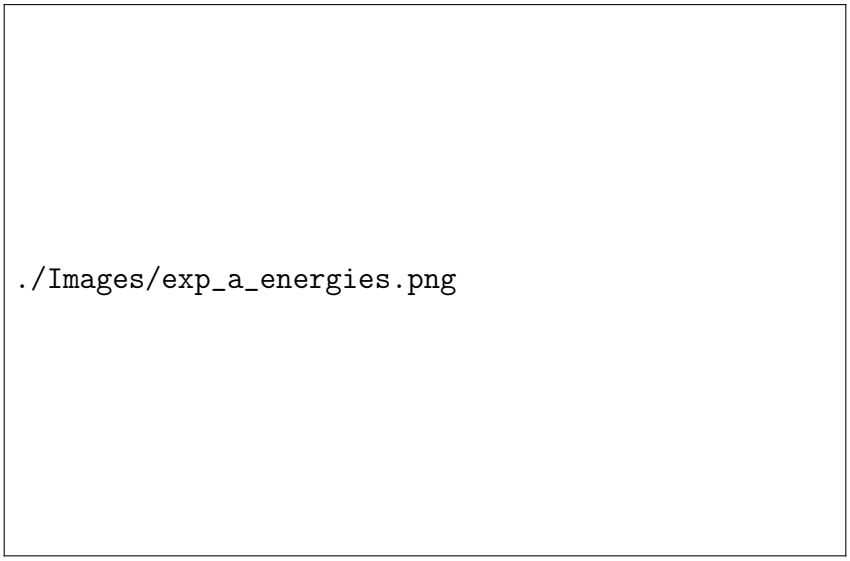


Figure 3: Dimensionless energies. Membrane (blue). Bending (green). Coupling (yellow). Membrane energy predicted by Kirchhoff-Love model $\varepsilon_{\mathrm{KL}} = \frac{a^4}{32\pi}$ (red). Plot in log-log scale.



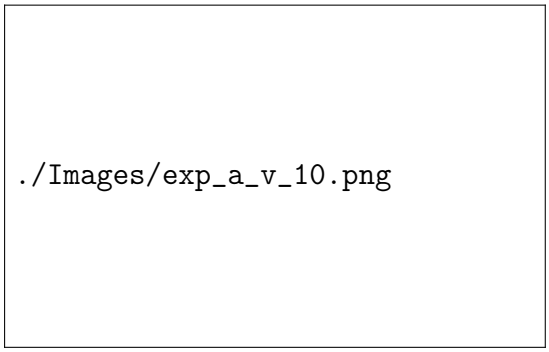
Figure 4: Dimensionless σ_{rr}.

observation explains why both the dimensionless bending and coupling energies decrease, as β and the relative disclination strength increases. Indeed, a plate subjected to a combination of a transverse load and a tensile membrane stress shows a smaller transverse deformation compared to the same plate free of the tensile stress (cf. [**?**][section 94, page 391-393]). By the same token, the dimensional bending energy decreases with β. In order to keep $\gamma = \overline{f}$, the dimensional load p_0 must scale with the inverse fourth power of h. Then when β increases at fixed R, h must decrease, and accordingly p_0. We also recall that, because for $p_0 = 0$, $w(x) = 0$ always solves the out-of-plane equation, when p_0 goes to zero the transverse displacement vanishes, alongside the dimensional bending energy. Nevertheless, the dimensional bending energy scales with $\frac{\mathsf{E}h^5}{R^2}$ (see section 0.2), which means that one cannot conclude that the dimensionless bending energy also decreases with h.

As a consequence, v in equation (**??**) is only activated by $\delta(0)$ and hence asymptotically approaches $v^0(x)$, solution to the problem ... [ALB] Finally, we numerically found that the dimensionless membrane, bending and coupling energies scale respectively with $\beta^{4.000}$, $\beta^{-1.738}$ and $\beta^{-1.169}$. In Figures 5 and 6 we report the profiles of, respectively, the

[ALB 12] unclear

dimensionless Airy's stress function $\tilde{v}$ and transverse displacement $\tilde{w}$ for $\beta = 10$ and $\beta = 100$. The profile of $\tilde{v}$ is self-similar and only scaled by the factor β^2. This confirms that the membrane equation is de facto decoupled from the bending problem. The trans-



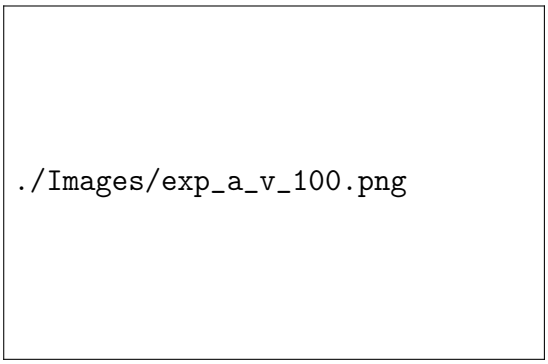
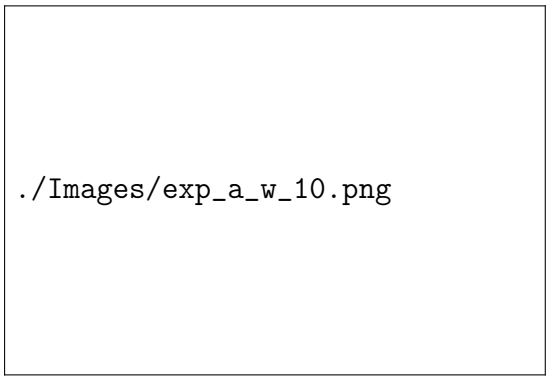


Figure 5: $\tilde{v}(\xi_1, 0)$ for $\xi_1 \in [-1, 1]$. LEFT: $a = 10$. RIGHT: $a = 100$.

verse deformation, conversely, shows a plateau in the middle of the plate and localised deformations akin to a boundary layer due to the clamped conditions, owing to the increasing intensity of the tensile stress field in the middle of the plate (see Fig. 4), as β increases.



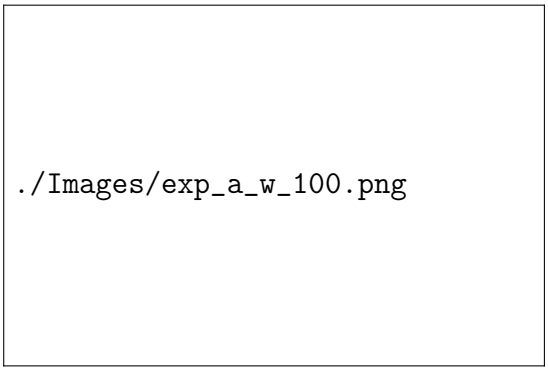


Figure 6: $\tilde{w}(\xi_1, 0)$ for $\xi_1 \in [-1, 1]$. LEFT: $a = 10$. RIGHT: $a = 100$.

The effect of the dimensionless external load γ When γ varies in the range $[0.1, 10]$ for fixed β, the total energy is essentially constant and matches the membrane energy predicted by the Kirchooff-Love model (see [**?**]), suggesting partial decoupling of the membrane equation in Eqn. (18) from the bending problem. We show in Figure 7 the dimensionless membrane, coupling, and bending energy components. The latter is quadratic with respect to the load multiplier, as inferred by inspecting the (strong form of the) bending problem considering the stress potential as a given. ALB

In Figure 9 we report the profile (as above, computed for $\xi_1 \in [-1, 1]$ and $\xi_2 = 0$) of $\tilde{v}$, which is identical to itself for any value of γ. In Figures 9-(1) and (2) we show the profile of $\tilde{w}$ for respectively, $\gamma = 0.1$ and $\gamma = 10$. Being, in this regime, the bending equation linear in the dimensionless load, the plot of $\tilde{w}$ is only scaled by γ, as expected.

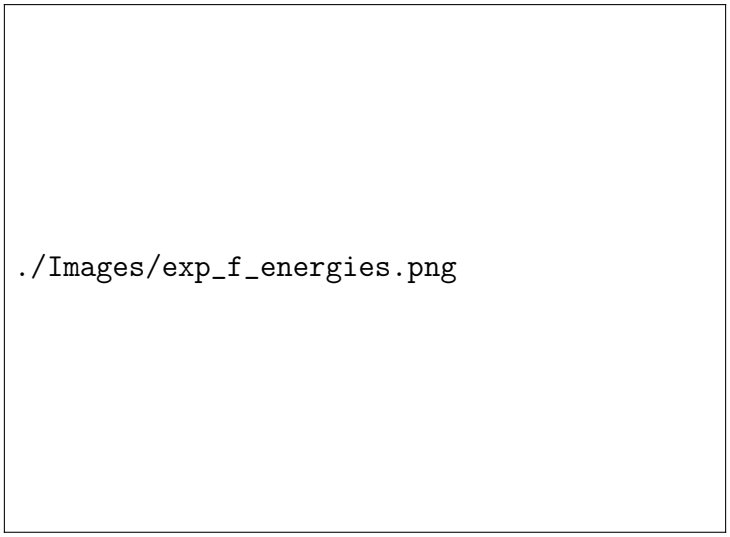


Figure 7: Dimensionless membrane energy (blue). Dimensionless bending energy (green). Dimensionless coupling energy (yellow). Membrane energy predicted by Kirchhoff-Love model (red).

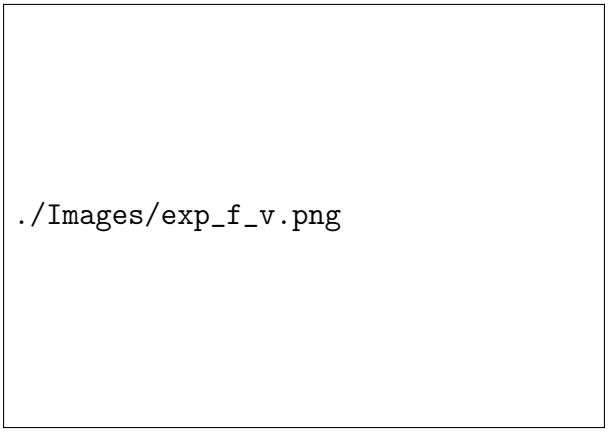
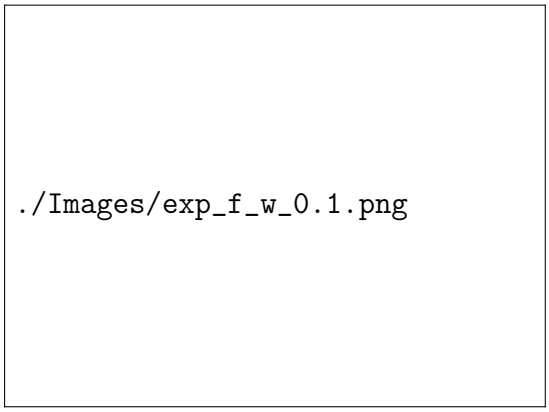


Figure 8: $\tilde{v}(\xi_1, 0)$ for $\xi_1 \in [-1, 1]$. LEFT: $a = 10$. RIGHT: $a = 100$.



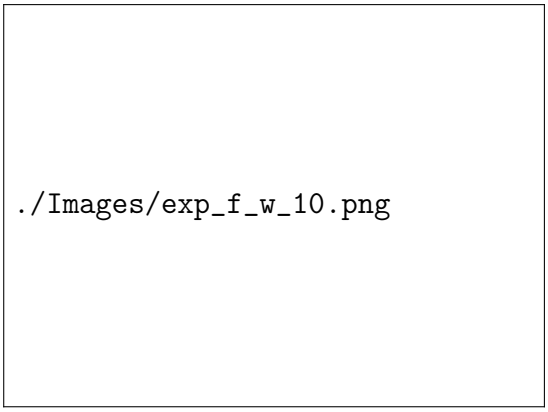


Figure 9: $\tilde{w}(\xi_1, 0)$ for $\xi_1 \in [-1, 1]$. LEFT: $\gamma = 0.1$. RIGHT: $\gamma = 10$.

1.3 Configuration

We focus now our attention to the effect that the sign of a disclination has on the membrane deformation. Let us first fix the setting. $\Omega \equiv B_1(0)$ ($B_1(0)$ is the unit ball centered on the origin). Then $R = 1$. We also set $E = 10\text{GPa}$, $\nu = 0.15$, $h = 0.05\text{m}$ and p_0 equals to the plate's weight per unit area ($p_0 := -\rho g h$ where $\rho = 10^3\text{kg/m}^3$ and $g = 9.81\text{m/s}$ is the gravity acceleration field). In the dimensionless setting we find: $\gamma = 0.016$ and $a = 20$. We further consider a discrete distribution of four disclinations of equal Frank's angle s to be specified later. Namely we set

$$\theta := s\big(\delta(x - t_1) + \delta(x - t_2) + \delta(x - t_3) + \delta(x - t_4)\big) \tag{25}$$

with $t_1 = (\frac{1}{2}, 0)$, $t_2 = (0, \frac{1}{2})$, $t_3 = (-\frac{1}{2}, 0)$, $t_4 = (0, -\frac{1}{2})$.

1.3.1 Case 1: negative Frank's angle s

We now take $s = -\frac{1}{2}$ in the definition of θ (see (25)). In Figures 10-(1) and (2) we show $\sigma_{rr} := \sigma e_r \cdot e_r$, with e_r being the radial unit vector. As already discussed in section 1.2, the stress field σ_{rr} is everywhere non-negative, which means it is of the tensile stress state. As a consequence the transverse displacement w is reduced with respect to the case $s = 0$

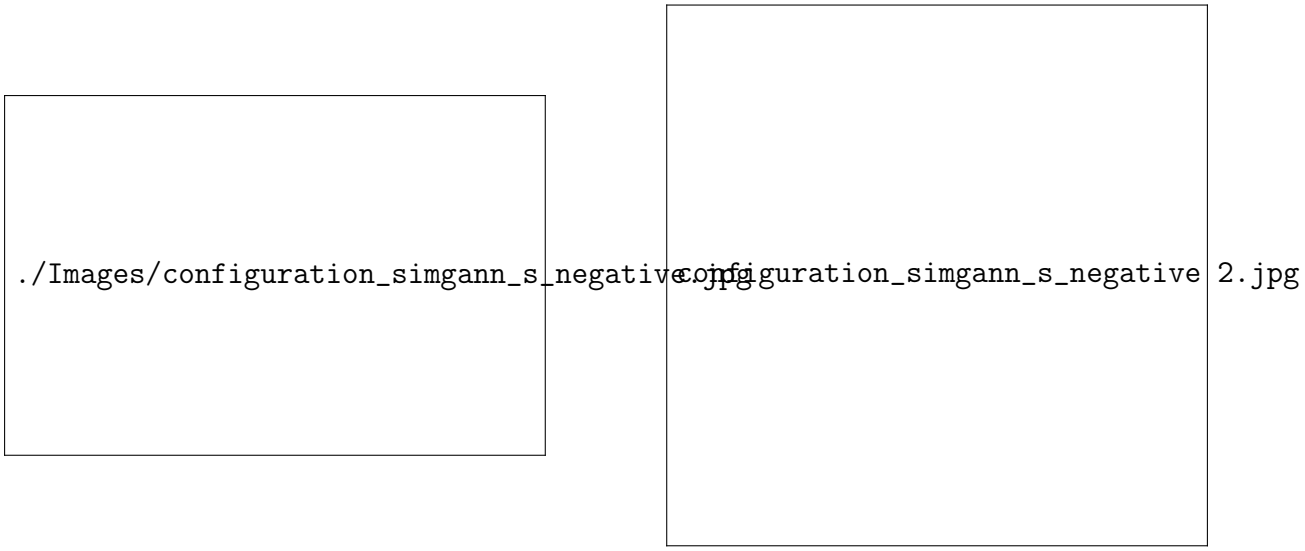


Figure 10: σ_{rr} [Pa×m] induced by a distribution of negative disclinations. Side and top views.

(see Figures 11-(1) and (2)).

1.3.2 Case 2: positive Frank's angle s

We turn now our attention to the case where $s = +\frac{1}{2}$ in (25). In Figures 12-(1) and (2), once again, we show σ_{rr} defined as above. σ_{rr} is everywhere non-positive, that is the plate is under a compressive stress state. In Figures 13 we show the profiles of the dimensional Airy's stress potential v and the dimensional transverse displacement w. The distribution of positive disclinations, has the effect to enhance the bending of the plate (compared to the case $s = 0$), and, moreover, induces the insurgence of of wrinkles [**?**] on the elastic surface. Phenomenon typical of plates subjected to a compressive external membrane

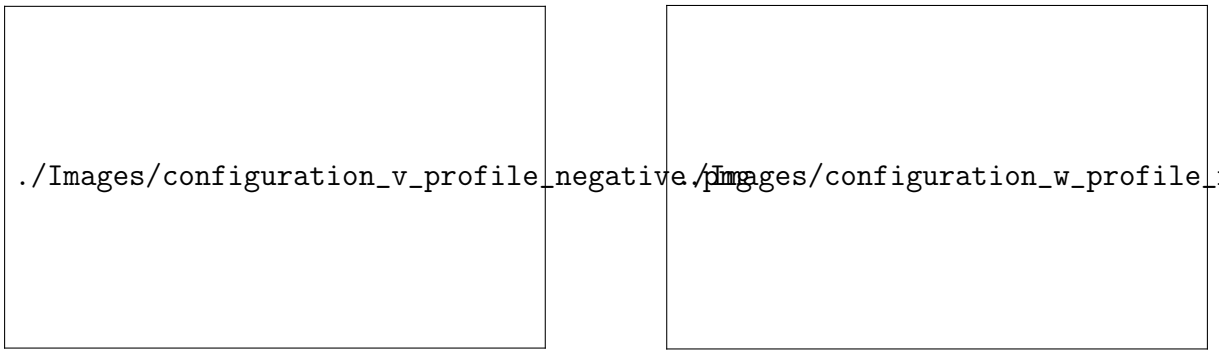


Figure 11: LEFT: v [Nm]. RIGHT: w [m]. Profile evaluated for $x_1 \in [-1, 1]$ and $x_2 = 0$.



Figure 12: σ_{rr} [Pa×m] induced by a distribution of positive disclinations. Side and top views.

load. However in this case the plate is completely free of any compressive external loads and this stress state is naturally induced by the presence of rotational defects in the crystalline lattice (the distribution of disclinations). Finally, in Figure 14 we report the

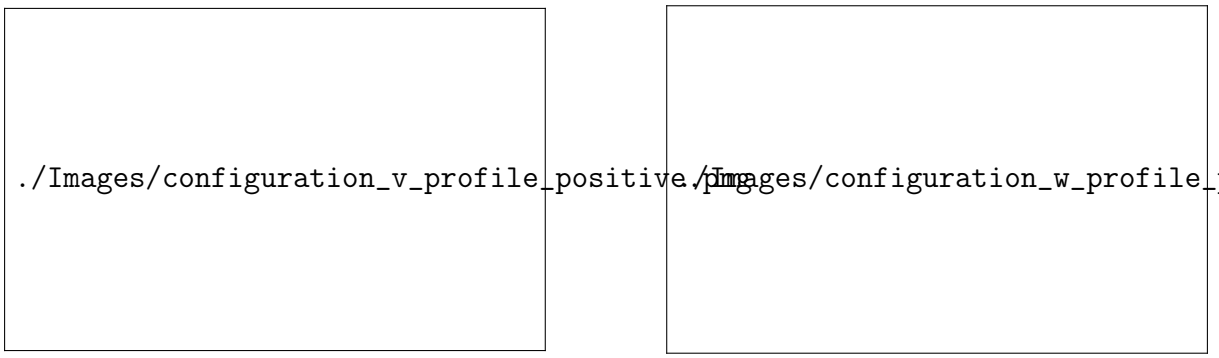


Figure 13: LEFT: v [Nm]. RIGHT: w [m]. Profiles evaluated for $x_1 \in [-1, 1]$ and $x_2 = 0$.

contour and the 3D plot of the plate elastic surface.

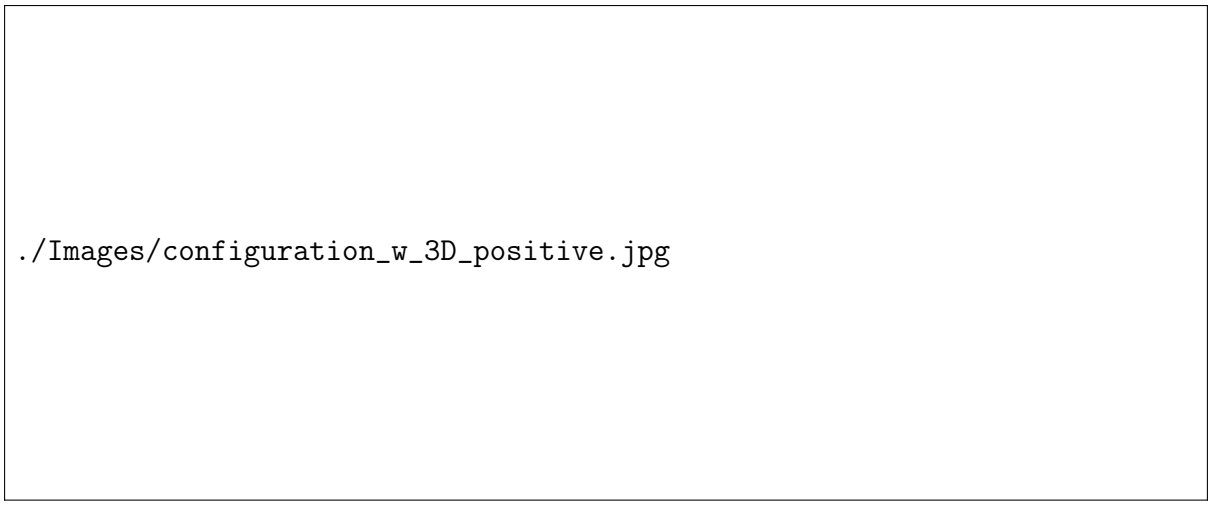


Figure 14: Wrinkling effect induced by the distribution of positive disclinations.